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1. Executive Summary

LinkPulse (working name) is a web-first progressive web app (PWA) that turns the lesson of the open-source Starlink monitor **Dishylink** into a universal product: a single, private, local-first dashboard for every network a household or small business touches. In markets like Kenya, a typical connected family already juggles Safaricom mobile data, a Zuku, Faiba, Airtel or Poa home fiber plan, opportunistic public WiFi and, increasingly, satellite options. Each of these comes with its own bundle logic, billing cycle, USSD menu and app. LinkPulse answers three questions no existing product answers together: **which network am I on, how is it actually performing, and what is it really costing me?**

The opportunity is grounded in measurable momentum. Kenya's fixed internet market grew 32.4 percent year-on-year to 2.84 million subscriptions by June 2026 according to regulator data, mobile money rails led by M-Pesa reach roughly 38 million monthly active users, and a wave of challenger ISPs is actively competing on price and quality. Meanwhile Starlink, the original inspiration for Dishylink, serves fewer than twenty thousand Kenyan subscribers. The monitoring problem that matters is therefore not satellite-specific: it is the fragmented, multi-operator reality of everyday connectivity.

2.84M Kenya fixed internet subs, Jun 2026 (CA data)	+32.4% fixed-market growth, year-on-year	37.9M M-Pesa monthly active users (H1 FY2026)	~19.5K Starlink Kenya subs, Sep 2025 (universality gap)
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This document makes the case that the Dishylink architecture pattern - on-device measurement, honest history, zero forced cloud, optional account linking for billing - can be generalized into a category-defining consumer utility for East Africa and, later, other multi-operator markets. It details the product concept, a staged and technically honest data-collection architecture, market sizing, an indicative four-stream revenue model, competitive positioning and an 18-month roadmap. We are seeking indicative pre-seed funding of USD 400,000 to ship the MVP, reach 50,000 monthly active users and open the first revenue streams within one year.

2. Origin: What Dishylink Proves

Dishylink is an open-source desktop and browser application, written primarily in TypeScript, that monitors the performance and health of a Starlink connection. It reads the dish and router directly over the local network, which means it keeps working during an outage - exactly when the user most needs answers. It renders live throughput, latency, power draw and obstruction telemetry; it records honest day, week and month history from its own measurements rather than from whatever the vendor's cloud chooses to expose; and it attributes per-device usage through router client counters. Connecting a Starlink account is strictly optional and unlocks plan and billing figures.

Three architectural decisions make Dishylink more than a hobby dashboard, and each one ports cleanly to a universal product. First, **local-first measurement**: the data lives on the user's machine, which builds trust, survives vendor API changes and costs the developer almost nothing in infrastructure. Second, **privacy as a feature**: no account, no cloud and no telemetry are required for the core experience, which is a genuine differentiator when the product must ask for sensitive permissions later. Third, **openness as distribution**: an MIT-licensed core with a browser extension turns satisfied users into installers and contributors, keeping acquisition costs near zero.

Dishylink capability	What it proves	LinkPulse generalization
Reads dish/router directly over LAN; works during outages	Measurement does not need the vendor cloud	Browser probes plus a thin Android shell plus an optional router agent, across any ISP
No account, no cloud, no telemetry; local storage	Privacy is a trust moat, not a constraint	Local-first IndexedDB; cloud sync is opt-in and end-to-end encrypted
Stat tiles; 15m/1h/6h and day/week/month history	Consumers adopt dashboards that stay honest	Same tile language applied to latency, outages and money
Per-device usage from router client counters	Household attribution matters on shared links	Per-device and per-network spend attribution
Optional account link for plan and billing	Billing context multiplies the value of telemetry	SMS, M-Pesa statement and e-invoice ingestion across operators
Outage and event logs with severity alerts	Evidence changes the consumer-ISP conversation	Timestamped evidence exports for complaints and refunds
Open-source core, desktop plus browser extension	OSS keeps acquisition cost near zero	OSS core plus installable PWA tuned for Android-first markets

Table 1: Generalizing the Dishylink pattern

3. The Problem: Fragmented Networks, Opaque Spend

East African connectivity is a portfolio, not a product. A Nairobi household may hold a monthly home fiber plan around KES 3,000-4,000, buy Safaricom data bundles weekly, hop onto office or campus WiFi by day, and top up an Airtel line when the primary network throttles. Small businesses stack a fiber line with a mobile failover. Each layer has its own price per gigabyte, expiry clock and failure mode, and the information about all of it is scattered across SMS confirmations, M-Pesa messages, USSD menus and carrier apps. No single screen tells the user what they spend, where it goes, or what they get.

Connectivity is not one product here - it is a portfolio of bundles, bills and workarounds, and nobody is keeping score.

- **Invisible spend.** Bundle purchases arrive as transient SMS notifications; by the end of the month the total is guesswork, and cost per gigabyte across networks is unknowable without a spreadsheet.
- **Unprovable quality.** When the line drops or crawls, users have no timestamped evidence, so complaints to ISP support desks collapse into their word against ours, and compensation clauses go unused.
- **Advertised versus realized speed.** Marketing quotes a headline number; nobody independently tracks what the household actually received at 8 p.m. in their own estate.
- **Blind switching.** Choosing between Safaricom, Airtel's newly aggressive fixed tiers, Zuku, Faiba, Poa or Mawingu happens on rumor, because there is no personal data to compare against.
- **No per-device truth.** Parents, landlords and SME owners cannot see which phone, TV or workstation is consuming the shared bundle.

The scale of the blind spot is growing, not shrinking. Fixed subscriptions in Kenya grew by roughly 700,000 lines in a single year while Airtel entered the fixed market with tiers from KES 1,999 and Zuku pushed discounted prepaid bundles; Uganda counts 44.3 million active mobile subscriptions and Rwanda passed 5.5 million internet users. More competition means more choices, more switching and more bookkeeping - precisely the conditions under which a trusted, neutral monitoring layer becomes valuable.

4. Product Concept: LinkPulse

LinkPulse is an installable progressive web app: one codebase that runs in the browser, installs to the Android home screen, works offline and stores its history locally in IndexedDB, inheriting Dishylink's privacy posture. The product is organized around three pillars that turn raw connectivity into decisions.

Pillar 1 - Measure

The app continuously runs lightweight latency and throughput probes, records online-offline transitions into an outage log, and offers on-demand speed tests. Users see the same honest stat-tile language Dishylink popularized - live downlink, latency, ping success and, on supported setups, per-device usage - generalized across every network the device touches. History is bucketed by day, week and month and is never averaged in a way that hides spikes, because the evidence value of the data depends on its honesty.

Pillar 2 - Track

Spend tracking meets users where billing already lives. In the installed experience, LinkPulse parses billing SMS notifications - bundle confirmations, expiry warnings, M-Pesa paybill receipts - using a community-maintained adapter registry per operator, and accepts imported M-Pesa statements or manual entries as fallbacks. Everything is normalized into cost per gigabyte, cost per day and per network, with alerts

before bundles expire or run dry. The output is a single monthly connectivity bill across all operators, something no carrier app will ever show honestly.

Pillar 3 - Act

Measurement and tracking convert into leverage. Users can export a timestamped outage evidence report when demanding refunds or SLA credits; consult a switching advisor that ranks the networks they actually use by realized quality and cost; and purchase bundles in-app through mobile money rails, with LinkPulse earning a commission. An opt-in, anonymized contribution builds the estate-level quality map that powers all three.

Persona	Context	Primary value
Urban household manager	Zuku fiber plus Safaricom data plus landlord WiFi	One bill view; bundle expiry and depletion alerts
SME owner (cafe, shop, clinic)	Business fiber with mobile hotspot failover	Downtime evidence for SLA claims; per-branch view
Campus student	Shared hostel WiFi plus daily bundles	Cost per GB; cheapest top-up path; speed checks
Remote worker / nomad	Fiber or Starlink with mobile failover	Outage log; failover guidance; proof for home-office SLAs

Table 2: Primary personas

5. Technical Feasibility and Architecture

The central engineering question is: how much truth can a PWA gather without fighting the platform? The honest answer defines a three-tier architecture, shipped in order of increasing capability and increasing permission sensitivity. Each tier is independently valuable, so the product is never blocked on a platform concession to deliver utility.

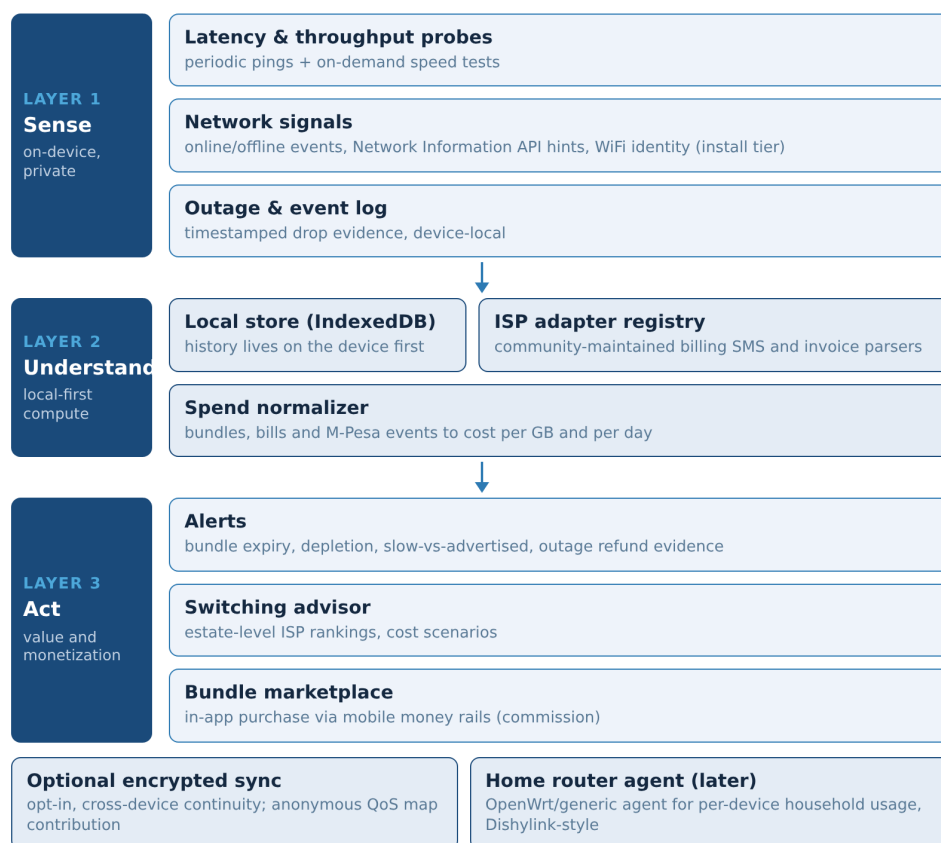


Figure 1: LinkPulse three-layer architecture - sense, understand, act, with optional sync and router agent

Tier	Capability set	Permissions and requirements	Data unlocked
Tier 1 - Browser core (PWA)	Latency and throughput probes; online/offline event log; Network Information API hints; on-demand speed tests; manual spend entries	Standard web APIs; installable PWA; no sensitive permissions	Connection-quality timeline; realized versus advertised speed; self-reported spend
Tier 2 - Installed intelligence	WiFi SSID/BSSID identity; SMS billing notification parsing; M-Pesa statement import	Android companion shell (Capacitor/TWA) with location and SMS-read permissions, user-consented	Which network am I on; what each network costs; cost per GB; bundle expiry alerts
Tier 3 - Network depth	Encrypted cross-device sync; anonymized QoS map; community ISP adapter registry; optional OpenWrt router agent	Opt-in consent for sync and map contribution; router-side install for the agent	Household per-device usage; estate-level ISP rankings; market-grade QoS corpus

Table 3: Data collection tiers

Tier 1 is unambiguous: modern browsers expose connection-change events, Network Information hints and the ability to run periodic probes against lightweight endpoints, which is sufficient for a credible MVP. Tier

2 acknowledges the real constraint - a browser cannot read SSIDs or SMS - and solves it the way successful African utilities (banking apps, Truecaller) already do: a thin, trusted companion shell that unlocks the sensitive sensors, wrapped around the same PWA codebase. Android's SMS and location permission model is mature, and the value exchange - your billing messages become your dashboard - is concrete and legible. Tier 3 mirrors Dishylink's router-direct philosophy for the household and creates the proprietary dataset: estate-level, per-operator realized-quality measurements that no incumbent owns.

Risks are managed by design rather than by hope. Permission policy drift is hedged by the tier ladder and by a SMS-forwarding fallback path; parsing fragility is contained by keeping the adapter registry community-owned, versioned and test-covered like Dishylink's firmware adapters; and privacy exposure is minimized by keeping raw SMS on-device, uploading only opt-in, aggregated, anonymized quality statistics.

6. Market: East Africa Connectivity

LinkPulse targets the East African Community core of Kenya, Uganda, Tanzania and Rwanda, beginning with Kenya as the design market. The region combines smartphone-led internet growth, hyper-competitive operator behavior and the world's most mature mobile money rails. Kenya alone counts roughly 2.84 million fixed internet subscriptions growing above 30 percent annually, a mobile market dominated by Safaricom's roughly 37.5 million connectivity customers, and an operator set - Airtel, Telkom, Faiba, Zuku, Poa, Mawingu, Starlink - that now competes aggressively on both price and quality.

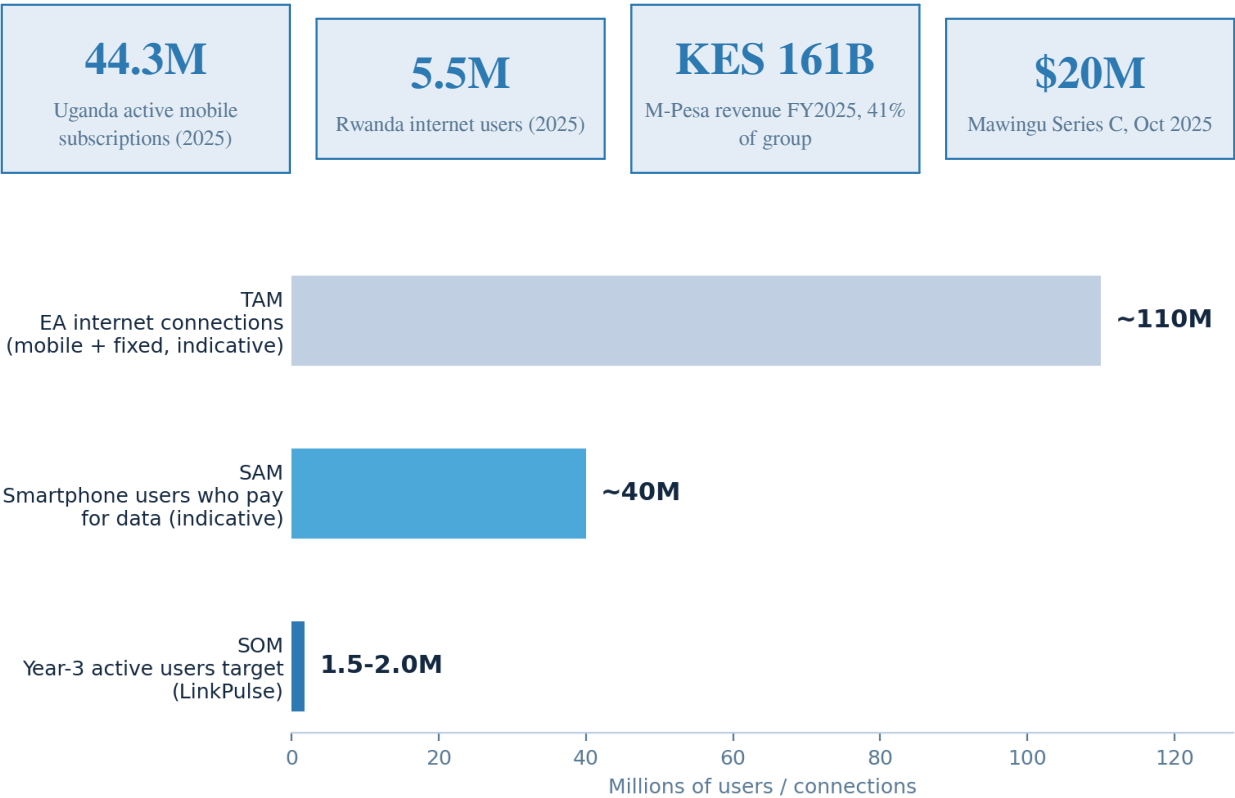


Figure 2: Indicative market funnel - TAM, SAM and Year-3 SOM (assumptions labeled indicative)

Sizing is intentionally conservative and labeled indicative. The top of the funnel - every internet connection in the four core markets - is on the order of 110 million mobile and fixed connections combined. The serviceable segment is the smartphone user who personally pays for data and manages at least one other connection, estimated around 40 million people. The Year-3 obtainable goal of 1.5-2.0 million monthly active users represents under five percent of that segment, achievable through the frictionless PWA distribution, open-source community channels and campus and estate ambassador programs described in the go-to-market plan. Even at conservative monetization (detailed in Chapter 8), that user base supports a multi-million-dollar annual revenue mix.

The investor context matters as much as the user math. Mawingu's USD 20 million Series C in October 2025 confirms deep capital appetite for East African connectivity; M-Pesa's 162 billion shilling annual revenue proves the payment rails on which commissions can flow; and the regulator's own publication of quality-of-service statistics legitimizes independent measurement as a public good.

7. Competition and Differentiation

Every adjacent player solves one slice of the problem; none combines cross-operator coverage with spend intelligence and independent quality evidence. Carrier apps are authoritative about their own billing but structurally cannot be neutral about a competitor's bundle or their own realized performance. Speedtest owns the one-shot measurement moment but keeps no honest history the user owns, and knows nothing about money. Network toolkits like Fing and GlassWire serve enthusiasts on desktops rather than households on Android. Truecaller demonstrates the pattern this product borrows: a permission-gated utility that converts a sensitive inbox into indispensable value.

Player	Category	Strength	Gap LinkPulse fills
MySafaricom app	Carrier self-care	Authoritative billing and payments for one operator	Single-carrier view; no independent quality evidence
Airtel / Telkom / Faiba apps	Carrier self-care	Per-operator bundles and support	Cannot compare rivals; no cross-ISP spend view
Ookla Speedtest	Measurement	Trusted one-shot speed test brand	No billing, no owned history, no advocacy
Fing / GlassWire	Network tooling	Deep device-level visibility	Desktop, enthusiast-oriented; zero spend intelligence
Truecaller	SMS utility	400M-user proof of permission-based value exchange	Not connectivity-focused; no measurement layer
Plan comparison sites	Content	Static awareness of tariffs	No personal usage data; no live quality signal

Table 4: Competitive landscape

Defensibility compounds along three axes. The **billing corpus** - anonymized, operator-normalized spend and quality measurements - grows more predictive with every user and cannot be retrofitted by an incumbent without admitting rivals' data into their apps. The **adapter registry**, maintained as open source, becomes the community's canonical parser library for East African billing formats, raising imitation costs. And the **consumer-advocate brand**, inherited from Dishylink's open, privacy-first DNA, is the one position a carrier-owned product can never occupy.

8. Business Model (Indicative)

The model layers four revenue streams on top of a single free utility, sequenced so that revenue begins before marketplace scale arrives. Pricing is indicative for the Kenyan launch market and deliberately anchored to what connectivity already costs users, not to Western SaaS benchmarks: a household spending KES 3,000-6,000 per month across networks will happily pay KES 100-200 for the tool that saves a bundle's worth of value every month.

Stream	Indicative pricing	Key assumption	Margin profile
Consumer Pro subscriptions	KES 100-200 per month	3-6% of monthly actives convert for history depth, family mode and evidence reports	High; digital delivery
Bundle and top-up commissions	3-7% of transaction value	In-app purchase share of tracked spend via mobile money rails	Medium; rail fees apply
SME multi-branch monitoring	KES 1,500-5,000 per month per business	1-2% of urban SMEs adopt SLA evidence and branch dashboards	High; seats expand per branch
Anonymized QoS insights	Licensing and report sales (opt-in corpus)	Operators and regulators value independent estate-level quality data	Very high; requires scale

Table 5: Revenue streams and indicative assumptions

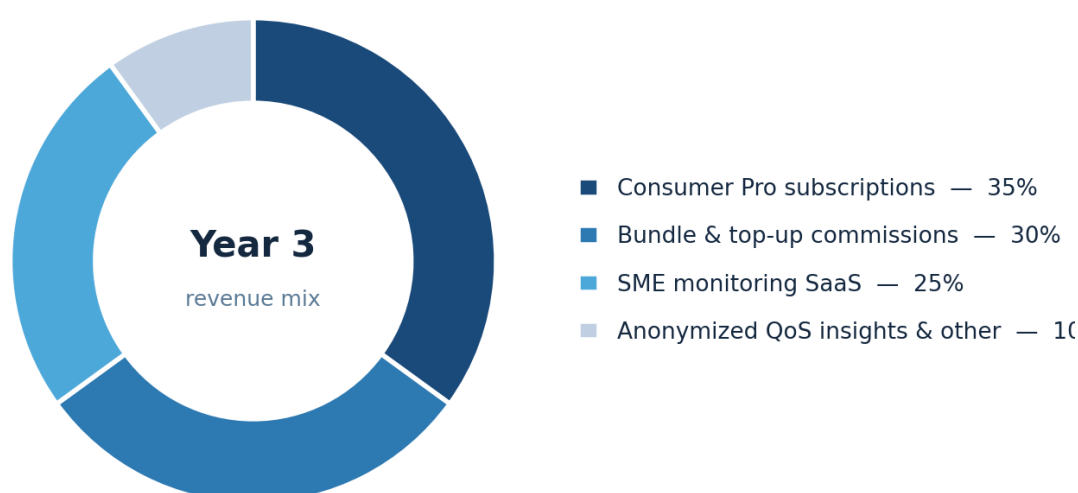


Figure 3: Indicative Year-3 revenue mix by stream (concept assumption)

Unit economics sketch as follows. Acquisition through the open-source community, PWA word-of-mouth and campus ambassadors carries a near-zero marginal CAC; the blended revenue per monthly active user across all four streams is indicatively KES 60-120 per month at maturity; and the cost base is dominated by a compact engineering team plus modest probe-endpoint infrastructure, since storage and computation live on user devices by design. The subscription and SME streams alone carry the company to sustainability; commissions and insights are the upside optionality that makes the category venture-scale.

9. Why Now, Go-to-Market and Investor Hooks

Three forces converge to make this the moment. First, competition has broken the incumbent's calm: Starlink's arrival and Airtel's fixed-market entry at KES 1,999 tiers triggered visible price moves from every operator, and consumers suddenly face real choices without real data. Second, the rails are ready: with 37.9 million M-Pesa users and mature smartphone penetration, both the commission revenue and the permission-based onboarding flow have working precedents. Third, the regulatory climate legitimizes the product: the Communications Authority already publishes quality-of-service statistics, and an independent, consumer-held measurement layer aligns with rather than fights the policy direction.

Go-to-market

- **Open-source core.** Ship the measurement engine and adapter registry on GitHub, Dishylink-style, converting developer credibility into free distribution and community-parsed operators.
- **Frictionless PWA entry.** No store gate for Tier 1; the install prompt and the Android companion shell arrive only when the user wants Tier 2 intelligence.

- **Estate and campus ambassadors.** Deploy in dense housing estates and universities where shared connections make the pain sharpest and word-of-mouth fastest.
- **Challenger ISP partnerships.** Poa, Mawingu and similar operators win when independent data proves their quality; early B2B pilots fund credibility before consumer scale.
- **Evidence-led virality.** Shareable monthly connectivity reports - what you spent, what you got - are designed to be posted, not just read.

Comparable trajectories

Truecaller scaled a permission-based SMS utility to roughly 400 million users and monetized identity on top of the inbox; Fing evolved from a free network scanner into a router-adjacent platform with device-security subscriptions; Ookla built a defensible measurement-data franchise that Ziff Davis acquired. Mawingu's USD 20 million round shows investors already underwrite East African connectivity infrastructure. LinkPulse sits at the intersection: consumer utility economics, infrastructure-grade data, and fintech-adjacent monetization on existing rails.

10. Roadmap, Risks and the Ask

The 18-month plan sequences capability, proof and monetization. Each phase has a kill-or-commit metric, and each phase's output is independently fundable evidence of traction. The architecture tiering from Chapter 5 maps directly onto this schedule: Tier 1 ships in months zero to three, Tier 2 intelligence in months four to six, and the marketplace plus expansion phases follow.

Phase	Months	Milestones	Success metric
Foundation	M0-3	PWA MVP: probes, outage log, honest history, manual spend entry, OSS repo public	5,000 installs; 40% week-4 retention
Intelligence	M4-6	Android shell beta: SSID identity, SMS billing parsing, M-Pesa import, expiry alerts	25,000 MAU; 1M billing events parsed
Marketplace	M7-9	In-app bundle purchase with commissions; switching advisor v1	50,000 MAU; first commission revenue
Expansion	M10-12	SME tier launch; Uganda entry; anonymized QoS map v1	150,000 MAU; 100 paying SMEs
Depth	M13-18	Tanzania and Rwanda; router agent; ISP insights API	400,000 MAU; breakeven trajectory

Table 6: 18-month roadmap

Risk	Likelihood	Impact	Mitigation
Platform permission policy changes (Android SMS/location)	Medium	High	Tier ladder decouples value from any single permission; SMS-forwarding fallback; router-agent path
Carrier resistance to independent measurement	Medium	Medium	Consumer-advocate positioning; insights product sold to, not weaponized against, operators
Copcats from incumbent apps	High	Medium	Cross-ISP corpus, adapter registry and OSS community compound faster than internal tools
Commission revenue needs scale	Medium	Medium	Subscription and SME revenue reach sustainability before marketplace maturity

Table 7: Principal risks and mitigations

The ask

We are raising an indicative USD 400,000 pre-seed to execute the Foundation and Intelligence phases: roughly 60 percent into engineering (a four-person team across PWA, Android and adapters), 25 percent into growth and community programs, and 15 percent into operations and compliance. The round buys 18 months of runway, the 50,000 MAU proof point, and the first two revenue streams live - the evidence base for a seed round priced on a category, not a feature.

11. Conclusion

Dishylink demonstrates that people will adopt a monitoring tool when it is honest, private and built at their own edge of the network. East Africa demonstrates that the need extends far beyond satellite dishes: a fast-growing, fiercely competitive, multi-operator connectivity market where consumers pay in shillings for quality they cannot see and spend they cannot total. LinkPulse connects those proofs with a deliberately staged architecture that earns the right to ask for sensitive permissions by delivering value without them first.

The wedge is narrow enough to execute with a small team and wide enough to matter: measurement first, spend intelligence second, marketplace and data products third. The rails, the competitive dynamics and the regulatory climate are aligned now, and the open-source heritage of the inspiration keeps trust - the scarcest asset in this category - at the center of the product. The invitation to investors is to fund the 12 months that convert a proven pattern into a category-defining utility for one hundred million connections.